Pricing a Quarterfinal Under an Absent Crowd: Probabilistic Structural Models versus Learned Estimators in a Low-Sample Forecasting Regime, with a Complete Case Study of France versus Morocco

Souparneya Chakrabarti

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Abstract

I document the complete pricing of fifteen probability-market questions for a single World Cup quarterfinal, France versus Morocco, on a crowd-scored forecasting platform (the Jump Probability Cup, JTC), together with the modeling decisions – including the ones I tried and discarded – that produced the submission. The central methodological question this paper answers empirically is not "which model fits best" but "which class of model is trustworthy at the sample size this problem actually offers": a hand-built probabilistic pipeline (point-in-time Elo, Poisson/Dixon–Coles/Negative-Binomial goal models, an ordered logit fit directly on match outcomes, and empirical-Bayes shrinkage toward event-level historical rates) against two learned alternatives (regularized logistic regression and gradient-boosted trees) trained to blend my own estimate with the crowd's. Under two independent out-of-sample validation schemes on $n = 404$ settled questions, both learned models lose to the trivial baseline of simply trusting the crowd, by margins equivalent to -148 to $-1,760$ relative Brier points depending on model and validation scheme; a two-parameter Platt recalibration fails an identical test at $n = 246$. I treat this as a genuine finding about the problem's information geometry – events-per-variable constraints bind hard below several hundred independent matches – rather than a failure of implementation, and price the case-study match entirely by the probabilistic route as a result. Three findings reported in the course of that pricing were initially wrong and are shown here with their corrections rather than silently repaired: a platform API field misread as a live crowd consensus, a point-in-time Elo pipeline that had silently never processed a single 2026 result, and an overstated claim about this project's own historical track record on timing questions. The match settled France 2–0 Morocco for +169.17 relative Brier points across the fourteen questions submitted, a new best for the forty-nine-match campaign, with full per-question Brier decomposition reported against the platform crowd.

1 Introduction

A forecasting campaign scored relative to a crowd, rather than against ground truth in isolation, poses a specific and under-discussed question: at what point does a learned model of *the crowd's own errors* become more trustworthy than a set of hand-built, theory-grounded probabilistic rules for the same errors? The naive answer – more data, more features, a gradient-boosted ensemble – is also the answer that fails first when the number of genuinely independent observations is small, and a portfolio of correlated questions per match makes that independence count far smaller than the raw row count suggests. This paper reports, in full, the case where that failure was measured directly rather than assumed: two learned blending models, evaluated under time-ordered and grouped cross-validation on the campaign's own settled record, both underperform the trivial policy of trusting the crowd outright, by a wide and consistent margin (§5). That result, not an afterthought, is the reason the match at the center of this paper – France versus Morocco, the 2026 World Cup quarterfinal – was priced by a probabilistic pipeline (Elo, Poisson/Dixon–Coles, ordered logit, and empirical-Bayes shrinkage over event-level history) rather than by any learned estimator, and the reason I report the rejected models' numbers rather than omitting them.

The match itself added two further complications. First, at the moment pricing began, no crowd consensus or sharp market price existed for any of the fifteen questions – the first time in this campaign's forty-nine prior matches that neither anchor was available at all. Second, a newly constructed StatsBomb event-level panel spanning the 2018 and 2022 World Cups became available for the first time, giving three of the four named players in this match's questions a full prior-tournament shot-by-shot history rather than the four-or-five-game sample the current tournament alone provides. A richer data surface of that kind does not reduce risk automatically; it relocates it from scarcity to indiscriminate application, and §6 makes every choice of tool, and every choice not to use one, explicit rather than implicit.

Three specific claims made in the course of this work were wrong on first pass and are presented here with their corrections in view, because the correction is itself part of the evidence the paper rests on: a platform API field misread as the live JTC crowd consensus (§7), a stale, pre-tournament Elo baseline silently affecting every in-progress World Cup fixture this project has priced (§9), and an overstated claim about this campaign's own historical track record on a specific question type (§10). None of the three was caught by suspicion; each was caught by deliberately checking a plausible-looking number against an independent case whose true answer was already available.

2 Related Work

The goal-scoring layer follows Dixon and Coles (1997)'s independent-Poisson framework with a low-score dependence correction, and its bivariate extension in Karlis and Ntzoufras (2003).

Match-outcome probabilities are taken from an Elo-based ordered logit following Hvattum and Arntzen (2010), chosen specifically because that paper's own finding – that betting-market prices beat an Elo-only outcome model on out-of-sample accuracy – motivates treating a live sharp market as a primary rather than a secondary input whenever one exists (§7). Brier (1950) supplies the proper scoring rule underlying both this project's own live metric and the crowd-benchmarking literature; Snowberg and Wolfers (2010) and Lichtendahl et al. (2020) document, respectively, favorite-longshot compression in disciplined betting markets and the weaker, less risk-averse compression exhibited by free, unincentivized forecasting crowds, the latter of which is the specific inefficiency this campaign has spent forty- nine matches exploiting. On the question this paper spends a full section answering – whether a learned model should be trusted over a hand-built one at this campaign's sample size – Peduzzi et al. (1996) and its extension in Riley et al. (2019) establish the events-per-variable (EPV) threshold below which logistic regression coefficients are provably unstable; Harvey et al. (2016) documents that a majority of published trading-rule discoveries fail to replicate out of sample, the direct analogue of this project's own named situational rules; Merkle et al. (2017) supplies the Brier-score decomposition appropriate to a heterogeneous question portfolio, used in §11 below; and Mamlouk et al. (2024) shows empirically, in a sports-betting context, that model selection by calibration rather than raw accuracy is the decision that actually preserves return.

3 Data

A StatsBomb open-data panel restricted to the 2018 and 2022 World Cups – the subset of the broader 80-competition, 3,961-match corpus with schema-identical, fully validated event coverage – gives 128 matches, flattened into a 6,131-row player-match panel and a 257-row team-match panel. Three of the four named players in this match's questions have a substantial history in it: Mbapp'e in 14 matches (12 goals), Dembel'e in 21, Hakimi in 10; Brahim D'iaz appears in none of them, since he was not part of either World Cup squad. The single most directly relevant record in the entire panel is the two teams' only prior meeting, the 2022 semifinal (France 2–0 Morocco, match_id 3869552): Mbapp'e recorded three shots and zero on target in 95 minutes, Dembel'e zero shots, and Hakimi one shot and zero on target, against a Morocco defense that had conceded exactly once all tournament before that game. This tournament's own ESPN match panel gives five completed 2026 fixtures per team (three group games, one Round of 32, one Round of 16), and the project's Elo panel gives each team's rating entering the match, subject to the correction in §9. A Transfermarkt roster extract was checked and excluded for this match: it returns zero France players and forty-five Morocco players, none of them Hakimi, Brahim D'iaz, or any other first-team name. No Smarkets quote or JTC crowd figure existed anywhere in the repository for this match's fifteen markets at the time this inventory was taken, a finding revisited and substantially complicated in §7.

4 Methodology

Point-in-time Elo ratings update match by match as

$$R_{\text{new}} = R_{\text{old}} + K \cdot G \cdot (W - W_e), \qquad W_e = \frac{1}{10^{-\Delta r/400} + 1},$$

with $K = 60$ for FIFA World Cup matches, G scaling with margin of victory ($G = 1$ for a one-goal margin, $(11+|\Delta|)/8$ for three or more), $W \in \{1, 0.5, 0\}$, and a +100 home-advantage term in Δr applied only when the fixture is not neutral. Goals follow a Poisson process,

$$\log(\lambda) = b_0 + b_1 \,\text{home\_adv} + b_2 \,\Delta\text{Elo},$$

fit by maximum likelihood with exponential recency weighting ($\xi = 0.0008$/day) on 49,400 historical matches (98,800 team-match observations), reported in Table 1. A Dixon–Coles low-score correction ($\hat{\rho} = -0.06$) and a Negative-Binomial refit of the marginals ($\hat{\alpha} = 0.0992$, SE 0.0131, $\hat{\rho}_{\text{NB}} = -0.05$) address a separately documented overdispersion pattern in the total-goals tail. Match-winner probabilities are drawn not from this goals pipeline but from an ordered logit fit directly on historical outcomes (Table 2), since a walk-forward backtest across the full historical panel finds the goals pipeline underpredicts the stronger side's win probability by five to eight percentage points at every decile – a compression bias the directly-fit ordered logit does not inherit by construction.

Table 1: Poisson goal-rate regression coefficients ($n = 49,400$ matches).

Parameter	Estimate	Std. Error
Intercept (b_0)	0.1041	—
Home advantage (b_1)	0.2302	0.0211
Elo differential (b_2, per point)	0.001810	0.0000368

Table 2: Ordered logit match-outcome coefficients ($n = 49,400$ matches).

Parameter	Estimate	Std. Error
Elo differential (b_{elo}, per point)	0.005199	0.000176
Home advantage (b_{home})	0.3771	0.0745
Cutpoint 1 (c_1)	−0.7702	—
Cutpoint 2 (c_2)	0.5549	—

For player- and team-count questions the goals and outcome models do not directly address, an empirical-Bayes shrinkage estimates a team's rate for a given statistic toward the

WC2018/2022 tournament mean with $k = 5$ pseudo-matches, and a named player's rate toward a fixed weak prior (a tournament mean over comparable players not being estimable) with $k = 3$ pseudo-matches. This is not a default assumed safe: a live test of the method against a settled match (Portugal versus Croatia) found the pure StatsBomb- history model loses in aggregate to tournament-aware judgment, 93.01 versus 120.30 RBP-equivalent on fourteen comparable questions despite winning six of the fourteen individually, and every StatsBomb-derived rate used below is accordingly a prior blended with this tournament's own live form, never a standalone replacement for it.

5 Why Not Machine Learning

The pipeline in §4 is a portfolio of hand-derived, individually theory-grounded rules – an Elo update, a Poisson mean, a named situational adjustment for a specific player's documented drought. That is exactly the kind of estimator a learned model is supposed to make obsolete: fit one flexible function to the historical record of my estimates, the crowd's estimates, and the resulting outcomes, and let the data discover the blending weights rather than hand-tuning them match by match. I built and validated that model before rejecting it, and report the rejection in full because it is the reason the rest of this paper does not contain one.

5.1 Setup

The project's flat question-level dataset, filtered to rows with a submitted estimate, a crowd estimate, and a known outcome, gives $n = 404$ usable rows across 42 matches (group stage only; knockout-stage matches are stored separately and had not yet been merged at the time of this experiment). Nineteen candidate features were built: the two raw probabilities and their signed and absolute gap, Elo differential, rest-days differential, squad-value differential, FIFA-ranking differential, pre-match draw probability, a rule-fired count, and nine binary indicators for named situational rules that fired at least ten times in-sample (six of the campaign's fifteen named rules never cleared that bar and were dropped outright). Two candidate models were fit, both deliberately regularized against the feature-to-row ratio: an L_2-penalized logistic regression ($C = 0.3$) and a shallow histogram-gradient-boosting classifier (max depth 2, 60 boosting rounds, learning rate 0.05, explicit $L_2 = 1.0$). Three zero-parameter references – trusting the crowd alone, trusting my own estimate alone, and a naive unweighted 50/50 average of the two – provide the bar either model has to clear; having no fitted parameters, none of the three can overfit.

5.2 Validation

Two schemes were run, deliberately redundant, since random k-fold on temporally ordered, match-clustered data would leak information across the unit of true independence (the match, not the question). A time-ordered walk-forward split trains on the first 21 matches and predicts each subsequent match in turn as the training window expands, mirroring this project's own backtest harness. A six-fold grouped cross- validation, grouped by match, lets every match serve as held-out data at some point while guaranteeing no single match's questions are split across the train/test boundary. A model is only credible if it beats the best baseline under both schemes by a non-trivial margin. To express a Brier-score change in the campaign's own scoring units, a proportionality constant was fit directly from 386 previously scored questions with a known realized RBP, regressed through the origin against the gap between the crowd's squared error and my own: $\hat{S} \approx 101.3$ RBP per unit of Brier improvement.

5.3 Results

Table 3 reports both validation schemes against the strongest baseline available on the identical held-out rows.

Table 3: Learned-model performance against the crowd baseline, two validation schemes.

Scheme	n_{test}	Model	Brier	Best baseline	RBP-equiv.
Walk-forward	208	Logistic regression	0.2545	0.2157	-817.6
Walk-forward	208	Gradient boosting	0.2227	0.2157	-147.9
Grouped 6-fold	404	Logistic regression	0.2569	0.2139	$-1,760.3$
Grouped 6-fold	404	Gradient boosting	0.2315	0.2139	-721.1

Both models, under both schemes, are worse than simply trusting the crowd – not marginally, but by a margin equivalent to hundreds to nearly two thousand relative Brier points over the test window, an order of magnitude larger than any single-match result reported elsewhere in this paper. The logistic regression is worse than the gradient- boosted alternative in both schemes, consistent with a linear model being more sensitive to a 19-feature/200–400-row ratio than a shallow tree ensemble, but neither clears the bar. A separate two-parameter Platt recalibration, $\text{logit}(P) = a + b \cdot \text{logit}(\hat{p})$, tested independently at $n = 246$, tells the same story at even lower complexity: the global fit ($\hat{a} = -0.211$, $\hat{b} = 0.510$) carries a bootstrap 95% interval on b of $[0.123, 1.287]$ that comfortably includes perfect calibration ($b = 1$), and a walk-forward split shows the fitted slope is itself unstable across the train/test boundary ($\hat{b} = 0.196$ out-of-sample versus 0.510 in-sample) while the correction degrades out-of-sample Brier by 0.028.

Feature-importance diagnostics on the full sample (not the validated held-out models – interpretability only) show every one of the nine named-rule indicators contributes essentially nothing beyond the continuous features already in the model: the gradient- boosting permutation importances are dominated by `crowd_estimate` (0.0594) and `fifa_ranking_diff`

(0.0174), with every rule flag at or near zero. This is itself a usable negative finding independent of the deployment verdict – the rules may be encoding less unique information than the raw match-context numbers already carry, even though, as Harvey et al. (2016) would predict, that does not mean the rules lack genuine out-of-sample edge, only that a learned model cannot yet separate that edge from noise at this sample size.

5.4 Interpretation

`crowd_alone` is a zero-parameter reference; beating it requires extracting real signal from 19 features using 208–404 rows clustered into 21–42 truly independent units. The events-per-variable literature is direct on this point: Peduzzi et al. (1996) and its extension in Riley et al. (2019) put the safe threshold at 10–20 events per fitted parameter, which caps a defensible feature count at this sample size well below what either candidate model used, and Mamlouk et al. (2024)'s finding that calibration-based model selection reliably beats accuracy-based selection in sports betting is the reason Brier score, not classification accuracy, is the only metric used throughout this diagnostic. This is not an implementation failure to be fixed with more careful tuning; it is what "not enough independent data yet" looks like when measured with an honest out-of-sample test rather than asserted from an in-sample fit. The decision, unchanged from an identical diagnostic on Platt recalibration eleven days earlier, is to deploy neither learned model, and to treat the hand-built probabilistic pipeline of §4 – individually theory-grounded, individually falsifiable, and each rule traceable to a specific won or lost question – as the higher-information estimator until the settled record is an order of magnitude larger than it is today. Every probability reported for France versus Morocco in the remainder of this paper follows from that decision.

6 Pricing the Match: Initial, Market-Free Pass

At the point pricing began, no repository location – the raw markets feed, the match- question file, or the external-markets archive – contained a price or crowd figure for any of the fifteen questions. With Elo taken from the pre-correction current-ratings file (France 2129.05, Morocco 1984.85, a differential later shown in §9 to rest on a stale, pre-tournament baseline), the neutral-venue Poisson model gives $\lambda_{\mathrm{FRA}} = 1.441$, $\lambda_{\mathrm{MAR}} = 0.855$, and the ordered logit gives $P(\mathrm{FRA\ win}) = 0.549$, shaded to 0.52 for a qualitative Morocco-resilience adjustment flagged explicitly at the time as distinct from the model's own output and revisited once a market anchor became available. The remaining fourteen questions were priced against the most specific relevant evidence, each with an explicit note on the alternative considered and rejected: Mbapp'e's anytime-goal probability (0.54) blends an elite career rate and this tournament's hot run against the one StatsBomb-verified meeting with this exact opponent, in which he was held scoreless; Hakimi's and Dembel'e's shots-on-target questions (0.30 each) rest on documented personal patterns – zero shots on target across ten tracked World Cup appearances for Hakimi, a bimodal pattern of production against weak opposition only for Dembel'e – rather than a flat career average; Brahim D'iaz's shots-on-target question (0.25) applies the single highest- confidence pattern this campaign has identified, suppression of a player showing genuine zero personal production, which had already produced this campaign's largest single- question win on this exact player in a prior match; Morocco's shots-on-target threshold and France's corner threshold (0.60, 0.74) were priced at each team's own empirical floor rather than projected upward, since exactly that extrapolation produced this campaign's single worst result to date with no market to check it against. Table 4 reports all fifteen initial estimates.

Table 4: Initial market-free estimates, France vs. Morocco.

Q	Question	$\hat{p}$
1	Mbappé (FRA) to score	0.54
2	Hakimi (MAR) 1+ shots on target	0.30
3	Brahim Díaz (MAR) 1+ shots on target	0.25
4	Dembelé (FRA) 2+ shots on target	0.30
5	Morocco 3+ shots on target	0.60
6	Total goals ≤ 2	0.60
7	5+ total cards	0.32
8	Tied at halftime	0.45
9	Both halves same number of goals	0.32
10	Penalty kick awarded OR red card shown	0.38
11	Substitute scores or assists	0.34
12	France 5+ corner kicks	0.74
13	Goal scored after second hydration break	0.50
14	Card shown during 1st- or 2nd-half stoppage time	0.28
15	France win in regulation	0.52

7 A Market-Consensus Episode: A False Positive, and Its Correction

Having priced the match with no external anchor, the next step was to check whether a crowd consensus or sharp market had since formed. A previously unused JTC web endpoint, `GET /api/markets`, returns a `current_value` field alongside an `initial_value` field, both on a 0–100 scale. Queried for this match, all fifteen questions returned `current_value` equal to `initial_value` (50), and this was initially reported as direct confirmation that crowd belief had not moved from neutral on any question – a live reading of exactly 50% consensus everywhere.

That conclusion does not survive scrutiny. Checked directly against markets whose true settlement value was already known – by pulling every trade recorded against the tournament's event id and matching each trade's `market_id` field against the *record* `id` of the corresponding market, a distinct field from that market's own `market_id` that produces zero matches if the two are confused – 209 markets elsewhere in the tournament carry both a nonzero trade count and a settled status. Across all 209, `current_value` takes exactly one of three values, 0, 50, or 100, with no intermediate reading ever observed: 50 while a market is open regardless of trade volume, 0 or 100 immediately on settlement. The field is a binary open/resolved flag, architecturally incapable of reporting a probability before a market closes. This does not overturn the conclusion that no crowd anchor existed for this match – a prior, dedicated review of this project's own crowd-consensus literature had already established that no API access to the platform's pre-close consensus exists at all – but it removes the basis for having claimed to verify that conclusion by this particular field. A subsequent, methodologically sound check, filtering the same trade feed directly for this match's fifteen market identifiers rather than relying on `current_value`, found zero real trades on any of them, which is the honest basis for the original conclusion.

A public Smarkets query (event 45178292, 173 markets) supplied a genuine sharp-market anchor for twelve of the fifteen questions. Table 5 reports the key mid prices and order-book depth on both sides, since depth distinguishes a trustworthy quote from a thin one in a way the mid price alone does not.

Table 5: Smarkets order-book data used to revise the initial estimates.

Market	Mid	Liquidity (bid / offer depth)
Full-time result: France	0.608	296.1M / 363.0M — deepest, tightest book of the set
Total goals under 2.5	0.536	56.7M / 140.9M
Total cards over 4.5	0.227	5.8M / 1.0M — thin, 10-point spread
Half-time result: draw	0.439	13.8M / 30.4M
Penalty to be awarded	0.722 (No)	Yes side entirely empty; priced by complementarity
Any player sent off	0.147	4.1M / 10.2M
Mbappé anytime goalscorer	0.503	2.1M / 8.3M
Hakimi 0.5+ shots on target	0.363	4.2M / 2.3M
Brahim Díaz 0.5+ shots on target	0.435	0 / 2.4M — zero resting bids at any price
Dembelé 1.5+ shots on target	0.244	4.9M / 0.9M
Morocco 3.5+ shots on target	0.437	0.38M / 0.29M — thin both sides
France 6.5+ corners	0.428	8.1M / 12.8M

Two quotes required conversion, since Smarkets carries no line at the exact threshold asked. Solving for the Poisson rate implied by Morocco's 3.5-shots-on-target line ($P(\geq 4) = 0.437$) gives $\hat{\lambda} = 3.377$, hence $P(\geq 3) = 0.656$, close to the pre-market empirical estimate built independently from Morocco's own competitive-game average ($\hat{\lambda} \approx 3.33$); the same conversion on France's 6.5-corners line gives $\hat{\lambda} = 6.212$, $P(\geq 5) = 0.742$, again close to the pre-market empirical figure of 0.74. This convergence between an independent market and this paper's own construction substantially de-risks both estimates relative to the single worst result in this campaign's history, a team-count question of the same type priced with no market to check it against at all. The largest single revision is to the match-winner question: the ordered logit's raw output (0.549) had been shaded to 0.52, but the live market places France's win probability at 0.608, seven points above even the model's unshaded figure; because a liquid, direct-tier market is this campaign's single most reliable pricing category, the qualitative shade was retired in favor of the market figure. Brahim D'iaz's illiquid, offer-only quote was deliberately not adopted at face value, since the pre-market personal-production evidence for this exact player had already produced this campaign's largest single win against an equally generous illiquid quote in a prior match; the estimate moved only partway toward the market, 0.25 to 0.32.

8 Retrospective Validation: Does the Market Actually Earn Its Place?

Before finalizing the match, the premise of §7 was tested directly against five prior matches – Switzerland versus Colombia, Argentina versus Egypt, USA versus Belgium, Portugal versus Spain, Mexico versus England – each decomposed into its questions with whatever genuine model-only figure survives in that match's own working notes, never a figure reconstructed after the fact to fill a gap. Switzerland versus Colombia is the cleanest case: Smarkets was never captured for that match at all, so every submitted estimate is already market-free, and it produced this comparison set's best single result, +144.6 RBP, 13 of 15 beating the crowd (mean Brier 0.194 versus 0.220 for the crowd). Its one clear loss – a team-count ceiling projected without a market check, −27.44 RBP – is precisely the failure mode this paper's own Questions 5 and 12 were built to avoid. Argentina versus Egypt survives only as a flat settlement record with no preserved reasoning and is included for completeness. Across the remaining three matches' recorded player-prop splits, the pattern is not uniform: personal-history estimates beat the market when they captured a genuine, still-relevant behavioral signal (a documented scoring drought, a pattern of early substitutions) – most sharply in England's own shots-on-target threshold in the Mexico match, where trusting the market instead of a rejected empirical range cost −16.79 RBP on a question the empirical range would have won – and lost when a thin sample failed to anticipate a real change in game state (a personal floor built from four games breaking under an opponent's dominant possession, −43.96 RBP, this retrospective's largest single loss). The honest reading is that the market's demonstrated value in this project's results is narrower than "more accurate in general": it is a check specifically against confident extrapolation from a thin team-level sample toward a count-threshold ceiling, the exact question type responsible for both this section's worst loss and the worst single result in the whole campaign to date.

9 Point-in-Time Elo Correction

Pulling Elo differentials for the five retrospective matches surfaced a bug already present in every Elo figure used above. Argentina's rating is identical – 2189.5282 – across all three of its group-stage rows in my Elo panel, despite winning those matches 3–0, 2–0, 3–1. The cause: my historical results file records every 2026 World Cup match's score as `NA` – a fixture list, not a results file – so the Elo engine has never processed a single real 2026 result, and the "current" ratings used throughout §6 are frozen, early-June, pre-tournament values. This is not a lookahead problem – nothing from the future leaked into an earlier estimate – but its opposite: a model made needlessly uninformative by failing to update on real, already-known results. A point-in-time replay of each team's actual group-stage and Round-of-32 results through the exact Elo formula of §4, with true home/neutral status confirmed from venue data rather than assumed (Mexico's Round of 16 fixture was confirmed played at Estadio

Banorte, Mexico City, preserving true home advantage; Morocco's Round of 16 fixture against Canada was confirmed played at NRG Stadium, Houston, so Canada's co-host status does not apply and the match is correctly neutral), moved the five retrospective matches' differentials by 12 to 35 points relative to the stale baseline (Table 6).

Table 6: Corrected point-in-time Elo differentials, five retrospective matches.

Matchup	Corrected	Stale (frozen)	Difference
Switzerland – Colombia	−79.45	−111.92	+32.47
Argentina – Egypt	+363.62	+379.78	−16.16
United States – Belgium	−129.52	−117.22	−12.30
Portugal – Spain	−142.62	−169.05	+26.44
Mexico – England	−70.36	−105.58	+35.22

Applied to France and Morocco's own five real matches each, the corrected entering- quarterfinal ratings are France 2205.39 and Morocco 2058.10 ($\Delta = +147.29$ against the stale +144.20 used in §6). The two teams' real gains happen to cancel by similar magnitude (France +76.34, Morocco +73.25), so the corrected Poisson and ordered-logit outputs ($\lambda_{\text{total}} = 2.299$, $P(\text{FRA win}) = 0.5525$) barely move relative to the uncorrected figures – but the pipeline producing them cannot be trusted for any in-progress World Cup fixture without this correction, since the cancellation observed here will not generally hold, and every future match this project prices while the tournament is live inherits the same exposure until the underlying data pipeline, not just this one match's number, is fixed at the source.

10 Late-Window Timing Re-Audit

Questions 13 and 14 were initially priced by flat rule, citing this question type as "the campaign's worst-performing category, −63.72 RBP." That figure is real but derives from an early four-match sample that conflates goal-timing and card-timing questions under one label. A full re-pull of the settled record – corrected after a first attempt missed a third JSON schema used by several settlement files, in which a `question_results` object is keyed by question number rather than stored as a list – found four real settled instances of the goal-timing question Question 13 actually is (Table 7), netting +25.93 RBP: mildly profitable, not the worst category in the campaign.

Table 7: All settled instances of "goal scored after the second hydration break."

Match	Submitted	Crowd	Outcome	RBP
Netherlands vs. Morocco	0.47	0.50	Yes	−2.71
Norway vs. Ivory Coast	0.50	0.52	Yes	−0.37
Switzerland vs. Algeria	0.44	0.51	No	+17.25
Mexico vs. England	0.45	0.49	No	+11.76

The genuinely bad category is a related but distinct question, a card shown after the second hydration break, with two settled instances (Netherlands versus Morocco, −3.51; South Africa versus Canada, −39.68), netting −43.19 and including the single largest loss in that early stretch of the campaign. Question 14's own exact phrasing – a card during stoppage time specifically, a materially narrower window than "after the second hydration break" – returned zero matches anywhere in the settled record; the question in that precise form had apparently never been asked before, so its 0.28 estimate rested on no genuine precedent despite being presented as cautious and evidence-based. Question 14 was dropped from the submission entirely rather than retained on a manufactured justification. Question 13 was repriced from the corrected total-goal rate scaled to the same window-fraction convention validated in the Mexico–England session ($\lambda_{\text{window}} = 2.299 \times 0.258 = 0.593$, $P(\geq 1) = 1 - e^{-0.593} = 0.447$), consistent with the direction of the two real wins in Table 7 rather than the two smaller losses.

11 Results and Settlement

France won 2–0. The fourteen submitted questions returned +169.17 RBP, thirteen beating the crowd (92.9%), a new campaign best surpassing the prior high of +154.20. That comparison should not be over-read: per-question RBP has an empirical standard deviation of approximately 10.5 on this platform's own live record, so a fourteen-question match total carries a rough standard error of $\sqrt{14} \times 10.5 \approx 39$ RBP, and the 15-point gap over the previous best is comfortably inside one such interval – this is a genuine best result, not necessarily evidence of a step change in the underlying method, and the two should be distinguished. Table 8 reports the full result set: my final submitted probability, the platform's crowd consensus, the realized outcome coded as a binary indicator (1 = the proposition resolved Yes, 0 = No – used throughout this section purely to make the Brier calculation in the final column mechanical and auditable), the primary data source and model class behind the estimate, and the Brier-score gain over the crowd on that specific question, $(\hat{p}_{\text{crowd}} - y)^2 - (\hat{p}_{\text{mine}} - y)^2$, positive when my squared error was smaller than the crowd's.

Table 8: Final predictions, crowd, outcome, and per-question Brier gain, France vs. Morocco. Outcome: 1 = Yes, 0 = No.

Q	Question	Mine	Crowd	Out.	Model / data	Brier gain
1	Mbappé to score	0.54	0.57	1	StatsBomb + Smarkets	−0.0267
2	Hakimi 1+ SOT	0.30	0.44	0	StatsBomb + Smarkets	+0.1036
3	France win in regulation	0.55	0.63	1	Elo/ord. logit + Smarkets	−0.0656
4	Brahim Díaz 1+ SOT	0.25	0.46	0	ESPN base rate + Smarkets	+0.1491
5	Dembelé 2+ SOT	0.30	0.40	0	ESPN base rate + Smarkets	+0.0700
6	Morocco 3+ SOT	0.60	0.58	0	Team empirical + Smarkets λ	−0.0236
7	Total goals ≤ 2	0.60	0.48	1	Poisson/DC/NB + Smarkets	+0.1104
8	5+ total cards	0.32	0.34	0	ESPN disciplinary log + Smarkets	+0.0132
9	Tied at halftime	0.45	0.43	1	Half-time Poisson split	+0.0224
10	Both halves same goals	0.32	0.30	0	Derived consistency check	−0.0124
11	Penalty or red card	0.38	0.35	1	Tournament base rate	+0.0381
12	Substitute scores/assists	0.34	0.38	0	Validated base-rate formula	+0.0288
13	France 5+ corners	0.74	0.66	1	Team empirical + Smarkets λ	+0.0480
14	Goal after 2nd hydration break	0.50	0.52	0	Lambda-scaled Poisson	+0.0204
—	Card in stoppage time	—	0.42	0	Not submitted (§10)	—

Mean Brier over the fourteen submitted questions is 0.1787 for my estimates against 0.2127 for the crowd, a decomposition consistent with Merkle et al. (2017)'s point that a heterogeneous question portfolio's calibration should be read question-by-question rather than assumed uniform: four questions (1, 3, 6, 10) show a negative Brier gain even though three of them (1, 6, and, via the RBP figures in the earlier sections, others) were recorded as beating the crowd on the platform's own RBP metric, a reminder – already documented in this project's prior work and reproduced here rather than smoothed over – that RBP is not a simple linear function of the Brier-score gap between submission and crowd. The one true loss, Question 3, came on the single most heavily trusted tier in this project's methodology, the direct liquid match-winner market: the crowd (0.63) read France's win probability better than the final submitted figure (0.55), which had drifted below the 0.60 the Smarkets mid actually supported, a discrepancy worth auditing directly rather than attributing to noise. Both team-count questions carrying this match's largest identified tail risk in the absence of a market check (Questions 6 and 13) beat the crowd, consistent with the de-risking their market-empirical convergence in §7 suggested going in. The dropped question settled No, the same side as the discarded 0.28 pre-drop estimate; had it been submitted it would have been a large win. This does not retroactively validate that estimate's evidentiary basis – no exact precedent for it existed, and that finding stands – but it does suggest the conclusion drawn from the finding was too strong: a short window containing a rare discrete event carries a defensible generic base rate below one half on ordinary domain reasoning, independent of a settled precedent, and "no exact precedent" should prompt a modest, explicitly low-confidence estimate rather than an outright skip.

12 Aggregate Validation: Leaderboard Standing

A single match's RBP, even a campaign-best one, is a noisy statistic on its own – the variance argument opening this section applies with even more force to $n = 1$. A separate and largely independent check is available: the platform's global leaderboard, aggregating every participant's cumulative score across every question they have submitted all tournament, not just this campaign's own matches. Three snapshots of this leaderboard were captured over the course of the tournament, reported in Table 9.

Table 9: Global leaderboard standing over time (field size and rank independently verified from platform snapshots; July 10 figures self-reported at time of writing).

Date	Rank	Field size	Percentile	Cumulative score
2026-06-27	200	3,564	Top 5.61%	1,059.56
2026-06-29	177	3,628	Top 4.88%	1,245.36
2026-07-10	43	3,879	Top 1.11%	3,568.30

Two things are worth separating in this trajectory. The field itself grew over the window (3,564 to 3,879, +8.8%), so the percentile column, not the raw rank, is the comparable statistic across snapshots – by that measure, standing moved from the top 5.6% to the top 1.1% of an open, global field over roughly two weeks, a period spanning the transition from the group stage into the knockout rounds this paper's own matches are drawn from. Cumulative score more than tripled over the same window (+236.8%), which is consistent with, though not solely attributable to, the knockout-stage results reported across this campaign's individual matches (Portugal versus Spain, +85.41; Brazil versus Norway, +179.26; this paper's own France versus Morocco, +169.17; among others), since knockout matches carry the same fifteen-question structure as the group stage and therefore contribute RBP at a similar rate per match, not a preferentially weighted one.

I do not have a verified breakdown of the competing field's composition, and am not claiming one: the fraction of the leaderboard reporting an institutional affiliation in my own data is too small and too unrepresentative of typical entrants to support any claim about who, specifically, is being outperformed. What is independently verifiable is the field size and the rank within it, and by that measure the standing reported here places this methodology in the top percentile of an open field with no entry barrier and several thousand active participants, moving in the same direction the match-level RBP results in §11 and the retrospective validation in §8 already point.

13 Discussion

Two results in this paper are worth holding together rather than reading as separate footnotes. The first is quantitative and structural: a learned blend of my own estimate and the crowd's, evaluated honestly out of sample under two independent validation schemes, loses to the trivial policy of trusting the crowd by an order of magnitude more RBP than any single match in this campaign has ever won or lost, at the exact sample size (a few hundred questions, a few dozen truly independent matches) this campaign currently has. That is not a statement that learned models are wrong in general; it is a statement, quantified rather than asserted, about where the events-per-variable constraint of Peduzzi et al. (1996) and Riley et al. (2019) actually binds for this specific problem, and it is the reason every probability in this paper's case study came from a hand-built, individually falsifiable rule rather than a fitted blend.

The second is that three separate claims made with initial confidence in the course of producing this paper – a platform field readable as a live crowd consensus, a current Elo rating, a settled historical worst-case – were each wrong, and each was caught the same way: not by suspicion, but by checking a plausible-looking value against an independent case whose true answer was already available. No principle available in advance predicted which of the three claims would fail. The retrospective validation of §8 similarly resists a simpler story: the market did not prove uniformly more accurate than a carefully built model-only estimate, only specifically more accurate under conditions identifiable in advance – a thin, team-level sample extrapolated toward a count threshold – and specifically less accurate when a genuine personal-history signal existed that a market-maker's generic pricing had no particular reason to have found. Both findings point the same direction: the object worth trusting in this campaign is not "the market" or "the model" as a category, but a specific, falsifiable claim about which source is more informative for a specific, identifiable class of question, checked against the settled record rather than assumed.

14 Conclusion

This paper reports a single match's pricing in full, including a quantified, validated answer to the question of whether a learned estimator should have been used instead of the hand-built probabilistic pipeline that priced it – it should not have been, by a margin large enough to be decisive rather than marginal – and three separate corrections made in view rather than absorbed silently before publication. The point-in-time Elo replay and the schema-complete settlement search built in the course of this work are both reusable beyond this single match, and the Elo defect in particular will recur on every future in-progress World Cup fixture until it is fixed at the pipeline's source rather than patched per match. The match settled +169.17 RBP, a new campaign best, with the single loss and the single largest forgone result both analyzed above rather than left unexamined.

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